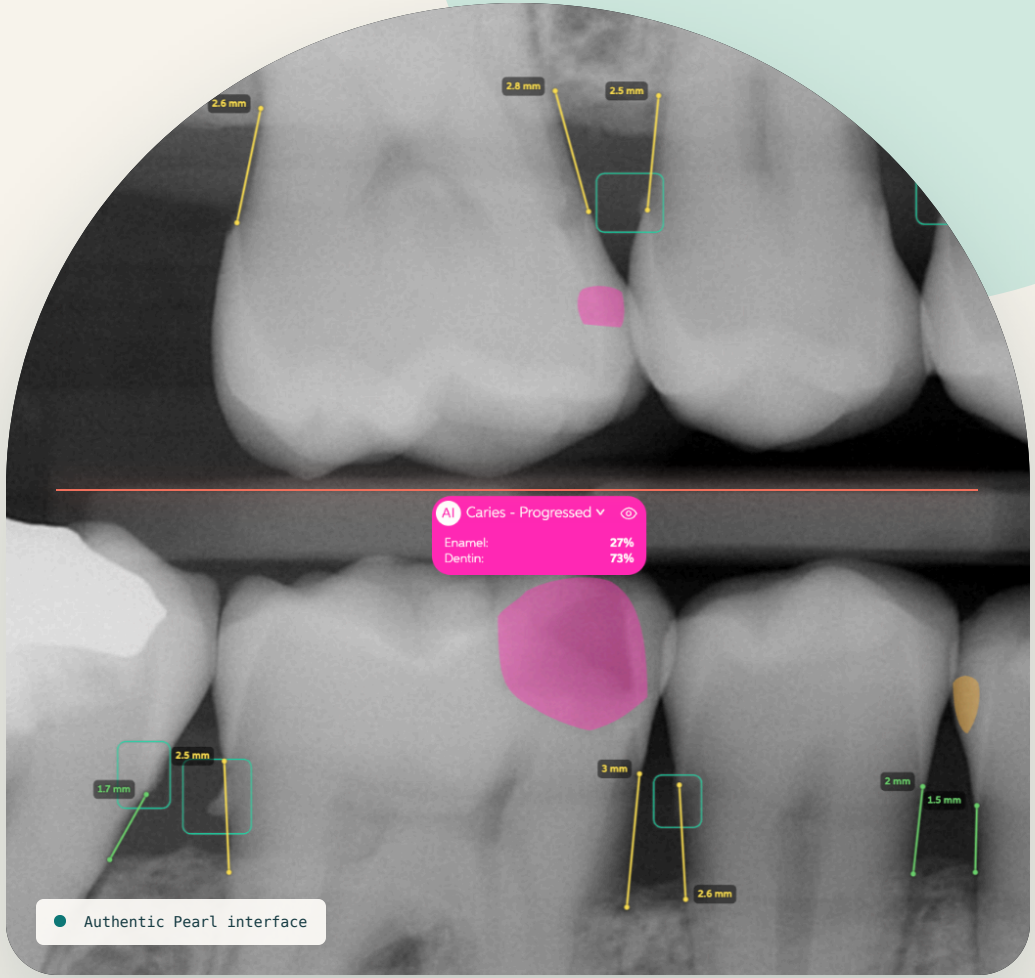
PEARL SECOND OPINION

Can AI see what the *human eye* misses?

AI-powered dental radiographic diagnosis, put on trial.

The British University in Egypt · AI Basics for Dentistry





The radiograph is clear. *The answer may not be.*

01 **Subtle lesions** Small density changes can be difficult to localise.	02 **Overlapping anatomy** Structures can conceal or imitate disease.
03 **Variable acquisition** Exposure, angulation and noise alter visibility.	04 **Human variability** Experience, fatigue and attention affect readings.

Potential beneficiaries: dentists, dental teams and patients—**if** the output is checked in context.

pearl®

SECOND READER / 01

A second set of eyes, *rendered in colour.*

Pearl Inc.'s **Second Opinion** is computer-aided detection software for dental radiographic review.

INPUT	Digital radiographs
ACTION	Detect + mark regions
ROLE	Decision support

It draws attention. It does not make the final diagnosis.

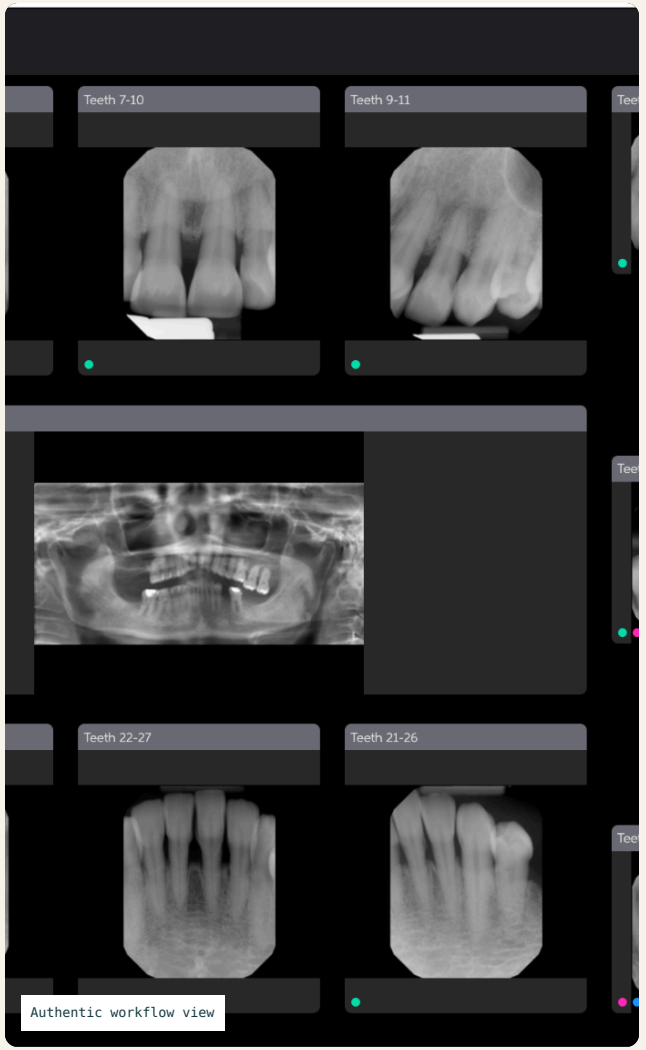


Official product image · Pearl User Guide

What goes into the system?

Scope changes over time. The label matters.

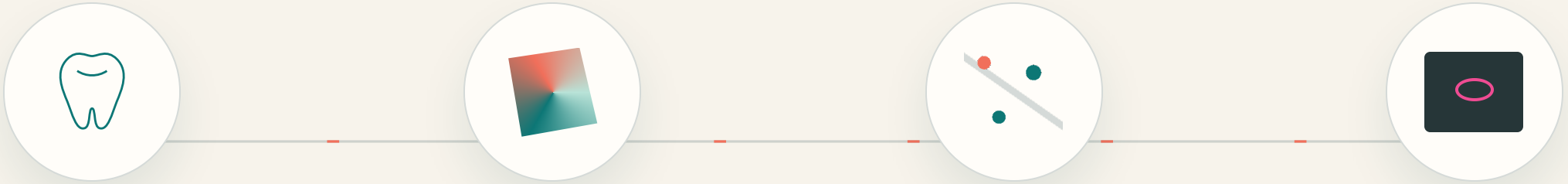
Product / clearance	Imaging input	Population + role
ORIGINAL · 2022 **Second Opinion** **K210365**	**Bitewing + periapical** Permanent teeth	**≥12 years** Second reader
LATER · 2025 **PR contours** **K242600**	**Periapical** Polygonal PR contours	**≥12 years** Second reader
LATER · 2025 **Second Opinion 3D** **K243989**	**Dental CBCT** Anatomy segmentation	**≥12 years** Concurrent + second reader



Authentic workflow view

Sources: FDA K210365 (2022), K242600 (2025), K243989 (2025). Later products ≠ original indication.

From pixels to *prompts for attention.*



01
IMAGE
Digital bitewing or periapical

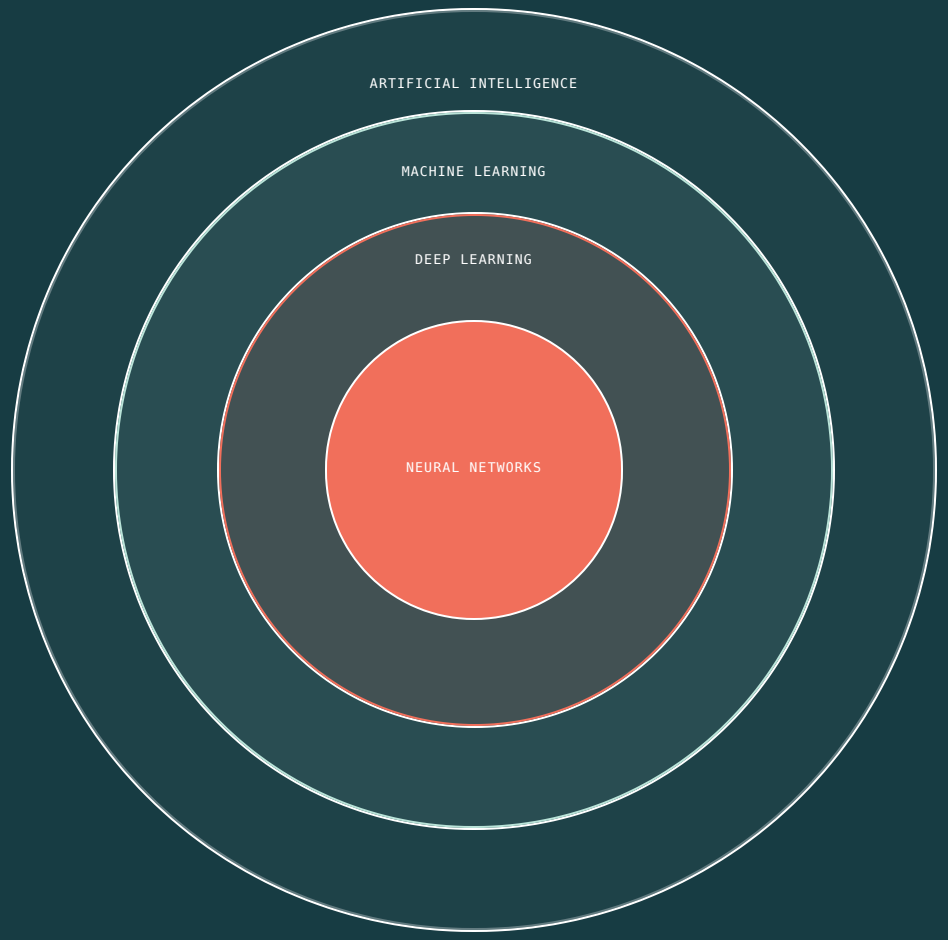
02
PROCESS
Pixels prepared for analysis

03
RECOGNISE
Learned visual patterns applied

04
DISPLAY
Metadata becomes overlays

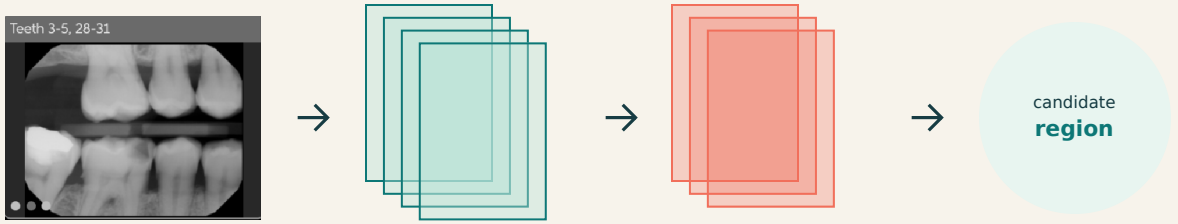
ESTABLISHED Cloud-connected computer vision + machine learning + graphical overlays.

Source: FDA K210365, Device Description. Diagram is an educational abstraction.



GENERAL EDUCATIONAL EXPLANATION

How a CNN reads an image



Convolution filters scan spatial patterns	**Layers** combine simple features	**Output** localises a candidate finding

DISCLOSURE LIMIT Exact proprietary architecture is not publicly documented.

A model learns from labels—*not clinical understanding.*

01
RADIOGRAPHS
Diverse de-identified examples

02
EXPERT LABELS
Regions + finding categories

03
TRAIN
Weights adjusted from error

04
VALIDATE
Tune without touching test data

05
TEST
Evaluate on unseen cases

GROUND TRUTH

The reference answer used to judge the model.

In K210365's clinical study: consensus of at least 3 of 4 expert readers.

PUBLICLY KNOWN
Machine-learning computer vision; neural-network-based algorithms; expert-consensus reference standard for the clinical evaluation.

NOT PUBLICLY DISCLOSED
Training-set size, exact composition, demographic balance, scanner distribution and proprietary architecture.

Clinical evaluation data ≠ model training data. Sources: FDA K210365; BUE Lecture 2.

Colour is a cue—
not a conclusion.

- **Incipient caries**
- **Progressed decay**
- **Calculus**
- **Notable margins**
- **Periapical radiolucency**

Toggle the overlay
Inspect the original image
Correlate clinically
Accept, reject or investigate

A detection localises attention. It does not establish disease.

ORIGINAL CLEARANCE · K210365

The 2022 evidence: *a multi-reader test.*

2,010	**25**	**370**	**655**
radiographs reviewed by all 4 GT experts	clinical readers in the MRMC study	caries lesion-containing radiographs	caries lesions in the FDA table

RETROSPECTIVE
multi-site, unblinded, open-label

→

GROUND TRUTH
≥3 of 4 experts agree

→

CROSSOVER READ
1,005 unaided + 1,005 aided

→

COMPARE
wAFROC-FOM

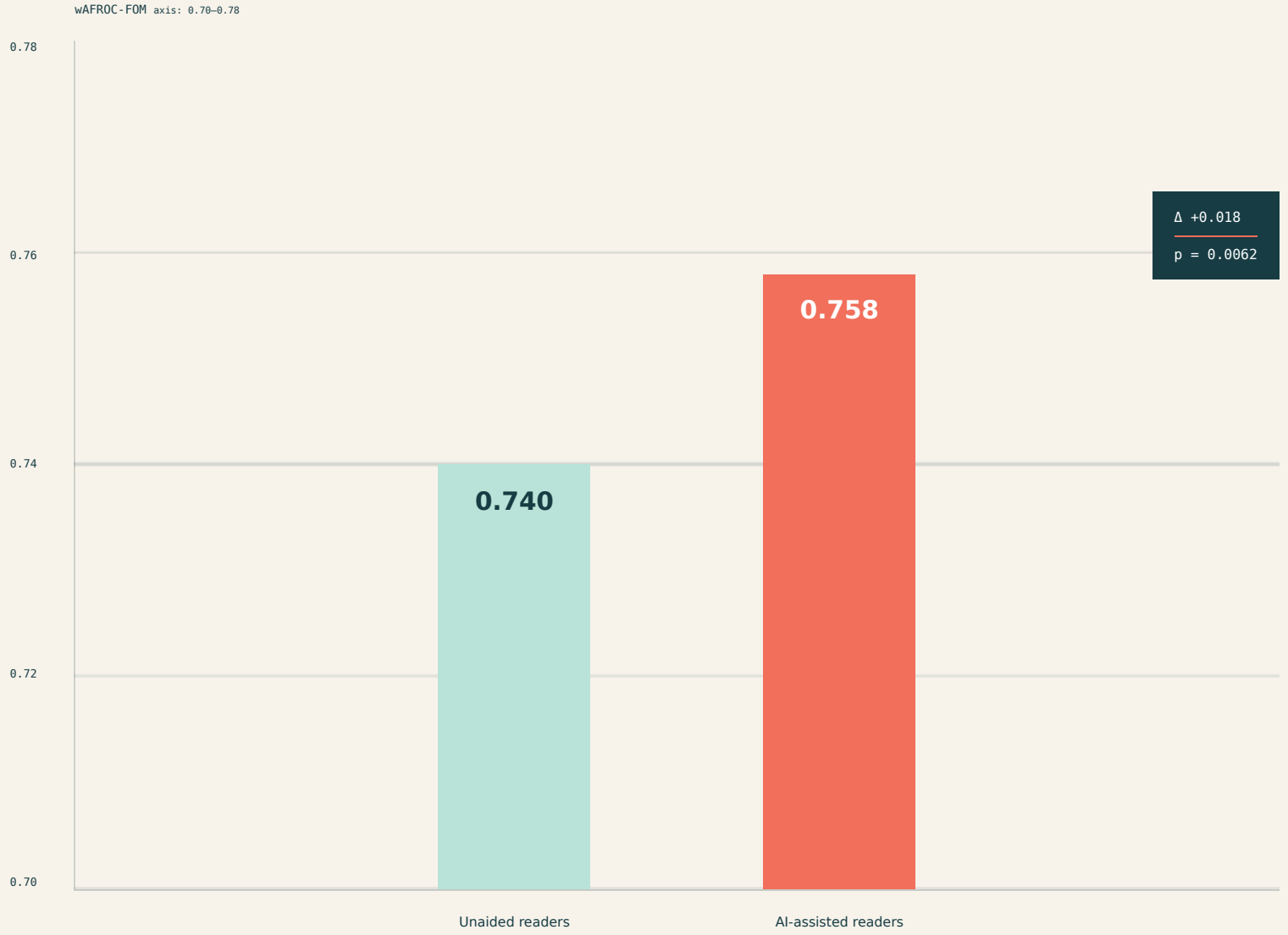
Accuracy note: FDA Table 2 reports 370 *lesion-containing radiographs*, not 370 total lesions.

Source: FDA K210365, Clinical Tests, Tables 2 and 6.

CARIES DETECTION · MRMC

A significant— *but modest—gain.*

wAFROC-FOM is a lesion-localisation performance figure of merit. It considers whether findings are correctly localised while penalising false marks.



NOT ORDINARY "% ACCURACY" Do not convert these figures into 74.0% and 75.8% diagnostic accuracy.

Source: FDA K210365, Table 6. Scale is truncated and explicitly labelled to make the small difference visible.

THE REGULATORY MAP · 2022–2025

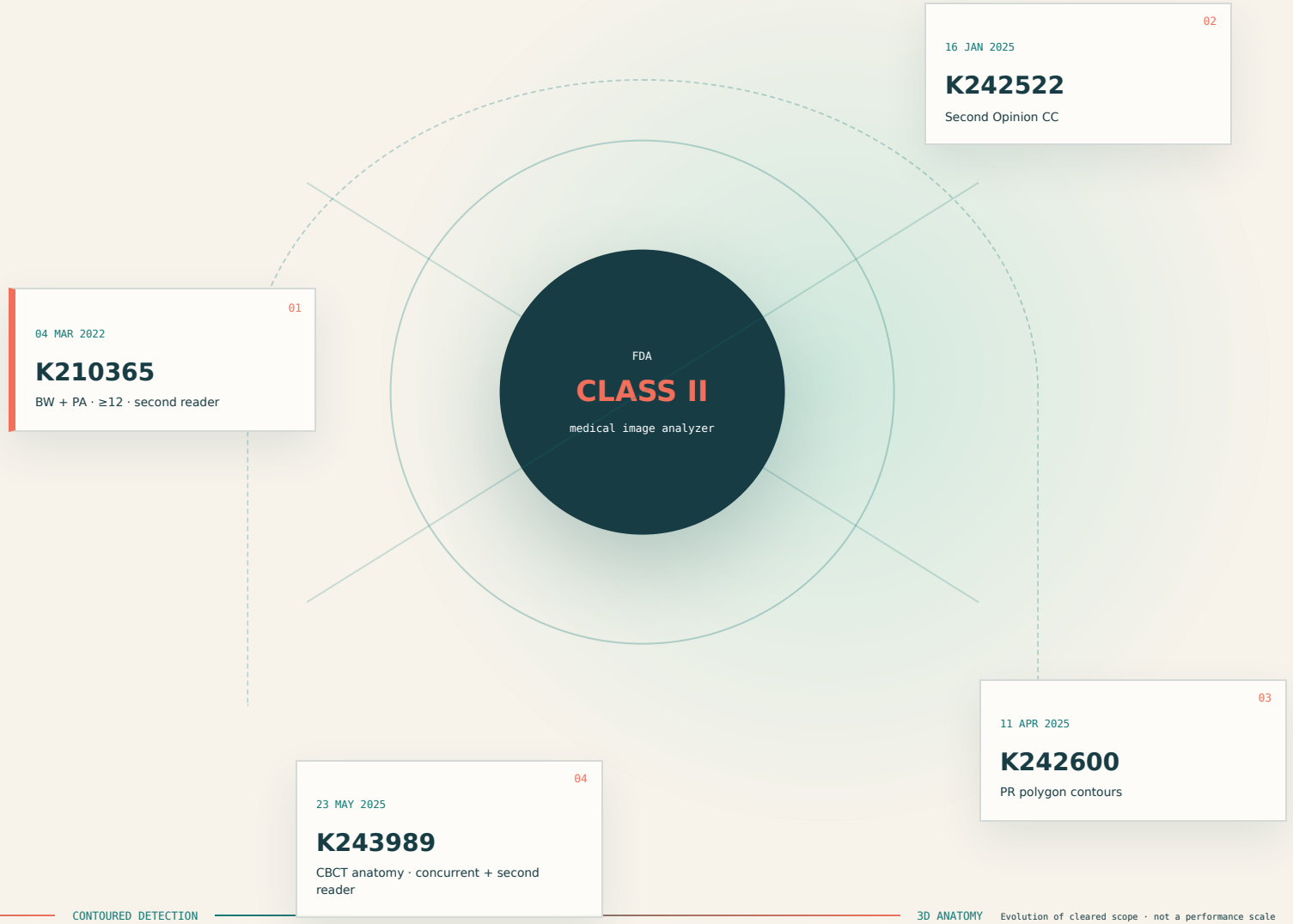
Clearance is a pathway— *not an endorsement.*

Each clearance has its own intended use. Later capabilities cannot be projected backwards onto the original product.

510(k)
substantial equivalence

≠

PMA
approval or endorsement



Sources: FDA clearance letters and database records for K210365, K242522, K242600, K243989.

A DETECTION IS A TEST RESULT

Every highlight creates *four possibilities.*



Accuracy can change with image quality, population, scanner and reference labels.

AI OUTPUT

REFERENCE / CLINICAL TRUTH	FLAGGED	NOT FLAGGED
FINDING PRESENT	TRUE POSITIVE **Finding flagged** Review + correlate	FALSE NEGATIVE **Finding missed** Risk: delayed care
FINDING ABSENT	FALSE POSITIVE **Healthy site flagged** Risk: unnecessary follow-up	TRUE NEGATIVE **Correctly clear** Still interpret in context

01 · IMAGE	02 · DOMAIN SHIFT	03 · AUTOMATION BIAS	04 · LABEL BIAS
Low contrast · positioning · artifacts	New patients · clinics · scanners	Trusting the overlay over contrary evidence	Experts and reference standards can disagree

RESPONSIBILITY FLOW

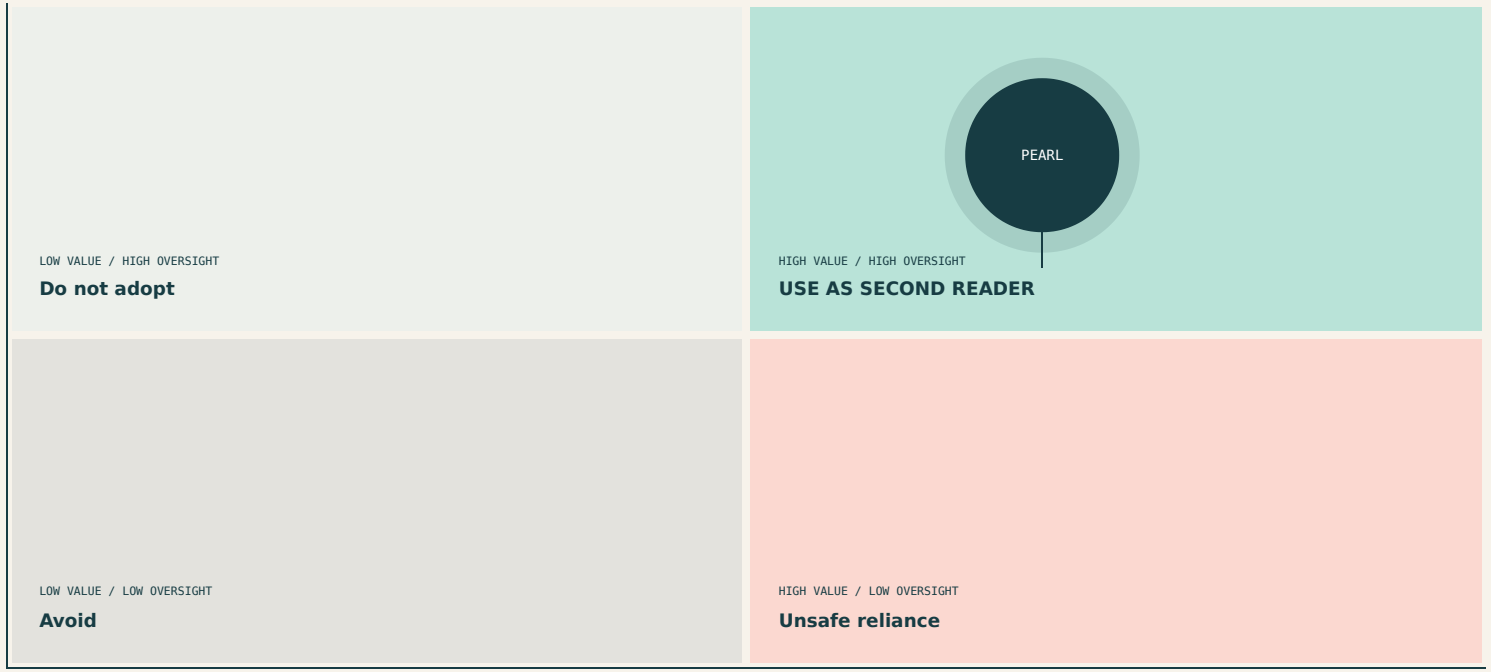
The AI informs. *The dentist decides.*



AI OUTPUT → CLINICAL INTERPRETATION → DENTIST ACCOUNTABILITY

Use it—
inside a disciplined protocol.

Our recommendation is conditional adoption: a second reader with high human oversight.





1. READ FIRST
2. CHECK IMAGE
3. TOGGLE AI
4. CORRELATE
5. DOCUMENT

THE NON-NEGOTIABLE

Read independently first. Toggle the AI second. Correlate clinically. Document the decision.



Artificial intelligence can improve how we see the radiograph.

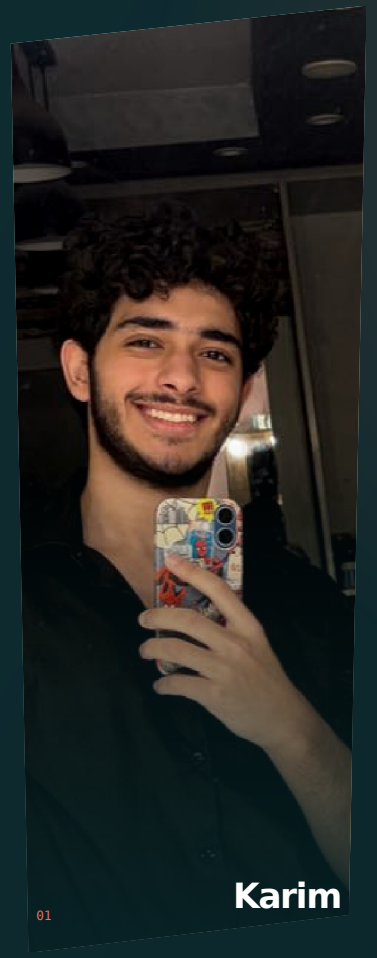
It cannot replace how we *understand the patient.*

DETECT. VERIFY. DECIDE.

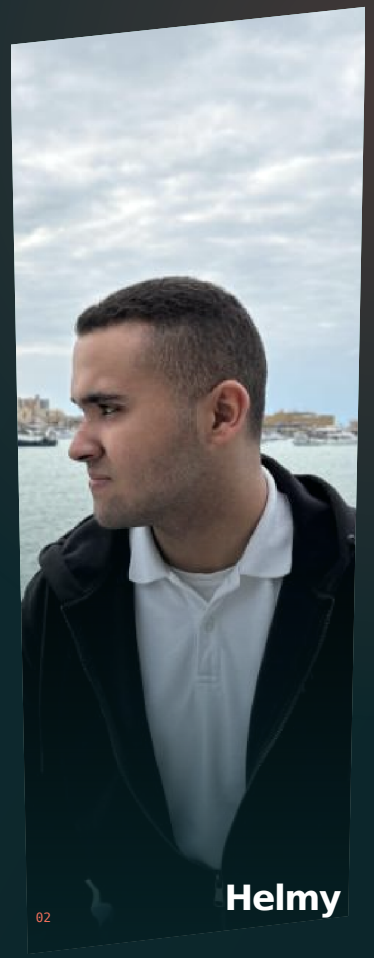
THE BRITISH UNIVERSITY IN EGYPT · AI BASICS FOR DENTISTRY

Four minds. *One clinical question.*

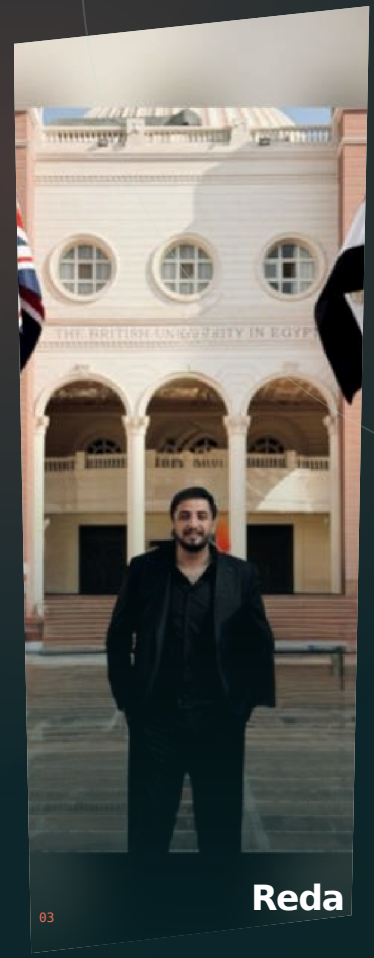
Who should make the final decision when artificial intelligence reads the image?



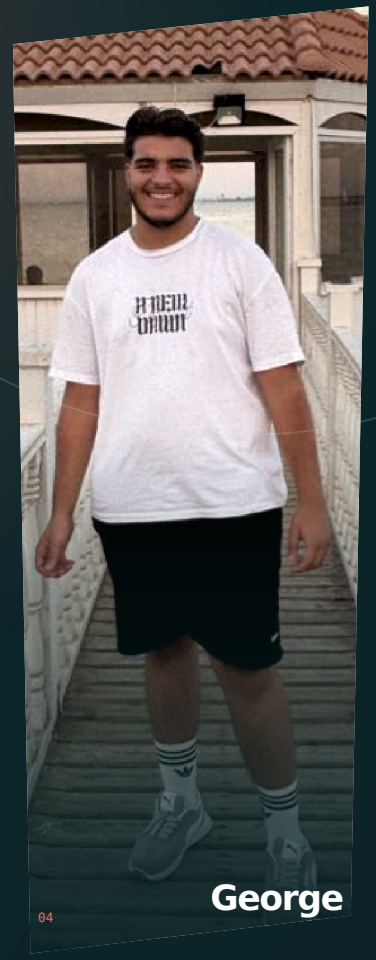
01 Karim



02 Helmy



03 Reda



04 George

SELECTED REFERENCES · FULL LINKS IN PDF METADATA

1 FDA K210365, Second Opinion®, 4 Mar 2022.

2 FDA K242522, Second Opinion CC, 16 Jan 2025.

3 FDA K242600, PR Contours, 11 Apr 2025.

4 FDA K243989, Second Opinion 3D, 23 May 2025.

5 Pearl Inc. Second Opinion User Guide.

6 Ibraheem et al. Diagnostics. 2025;15:1432.

7 Ji et al. J Dent Educ. 2026;90:1342–49.

8 BUE AI Basics for Dentistry, Lectures 2–3.